

EMMANUEL ABLAO

Full Stack Developer

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PROFILE

Full-Stack Developer with hands-on experience building web, mobile, desktop, and IoT-integrated systems. Proficient in React, TypeScript, Node.js, and SQL/Supabase, with experience deploying production applications (SSL/DNS configuration, Hostinger, GoDaddy) and integrating embedded hardware (ESP32, NFC) with backend APIs for real-time data processing. Adaptable and self-driven, with a continuous focus on learning modern web technologies and best practices.

SKILLS

Frontend

React • JavaScript • TypeScript • React Native

Backend & Database

Node.js • SQL • Supabase • Firebase

Desktop

Electron

IoT & Embedded

Internet of Things (IoT) • Arduino • PlatformIO

Tools & Version Control

Git • NPM

PROJECT EXPERIENCE

Inventory Control System

Philippines

- Built full-stack inventory management application using React, Vite, and TailwindCSS with a Supabase (PostgreSQL) backend for centralized, secure data management.
- Implemented offline-first functionality using Service Workers and IndexedDB, enabling uninterrupted stock scanning and purchase order creation with automatic sync on reconnect.
- Developed secure authentication flow using Supabase Auth, protecting access tokens and user data.
- Built a real-time synced dashboard with Recharts, using Supabase real-time subscriptions to keep stock levels and alerts updated across all active sessions.
- Implemented multi-role access control (Admin, Manager, Staff) with Row Level Security (RLS), audit trails, automated purchase order generation, and CSV reporting.

EAT-ALL POS

Philippines

- Developed a cross-platform desktop POS application using Electron and React, supporting Windows, macOS, and Linux.
- Built an offline-first architecture using SQLite for local data persistence, managed via IPC bridges between main and renderer processes.
- Designed a touch-optimized cashier interface with real-time VAT and discount calculations, multiple payment methods, and quick product search.
- Implemented real-time inventory tracking with automatic stock deduction on order completion.
- Built an analytics dashboard delivering daily and weekly sales insights, top-selling items, and low-stock alerts.

Quick Track

Philippines

- Built a cross-platform mobile application for field technicians using React Native and Expo (iOS and Android).

- Implemented offline-first data syncing with local SQLite queueing, batched REST API uploads, and per-record retry logic.
- Developed equipment management features covering multiple categories (HVAC, Electrical, Plumbing, Safety, Mechanical) with serial number tracking and service logs.
- Built on-device reporting with camera integration and shareable PDF report generation.
- Implemented a push notification system for overdue equipment alerts and scheduled daily review reminders.

Vue sur la Montagne

Philippines

- Developed a fully responsive single-page React application for a resort booking platform.
- Built custom animation hooks using Intersection Observer for scroll-based interactions and smooth page transitions.
- Designed a hero carousel with touch and keyboard navigation, integrated with a live booking form including date pickers and guest selectors.
- Built an interactive accommodations showcase with hover details, an amenities grid, and an animated testimonials gallery with lightbox.
- Implemented a validated contact form for guest inquiries.

ERJ Smart Solutions

Philippines

- Built an HR and attendance management platform supporting mobile app clock-in and dedicated desktop terminal clock-in, with real-time sync to a central system.
- Integrated custom hardware (ESP32 with PN532 NFC module) for physical attendance scanning, transmitting real-time data to the backend for processing.
- Designed a scalable, subscription-based architecture supporting multiple company accounts with tiered feature access.
- Built a centralized dashboard for department management, shift coordination, attendance tracking, and leave request handling.
- Implemented role-based access control (Admin, HR, Manager, Staff) with strict permission enforcement to protect sensitive workforce data.

JJC Engineering — SAP System

Philippines

- Designed and developed a full-stack SAP-style enterprise platform for JJC Engineering, replacing manual Excel-based workflows for payroll, attendance logging, and total hours computation across all departments.
- Built a responsive web application with a public landing page and a secured employee dashboard featuring Announcements, Attendance time-in/out, a Cooperative Savings Program (Coop), and a Personal Information portal displaying employee benefits.
- Implemented a Department Selector routing system with dedicated modules for HR, Finance, Procurement, and Operations, enabling role-specific access and workflows per department.
- Secured all API communications using JWT-based authentication with AES-256-CBC symmetric encryption on all frontend requests and backend responses, protecting sensitive employee and payroll data end-to-end.
- Developed a companion cross-platform mobile application (React Native) mirroring all web features, with Firebase Cloud Messaging (FCM) push notifications to alert employees of new announcements and updates even when the app is closed.
- Built an offline-first desktop kiosk application for barcode-based employee time-in/out at building entrances, queuing attendance records locally when offline and automatically syncing to the server upon internet restoration.

EDUCATION

Bachelor of Science in Information Technology

Antipolo City, Rizal | Sept 2022 – Dec 2025

ICCT Colleges — Awarded Best in Website Development

BS Information Technology (Vocational, 2 yrs)

Antipolo City, Rizal | May 2016 – Oct 2017

Gateway Institute of Science and Information Technology